BARCS: beta-binomial regression for multivariable CRISPR screen designs

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Abstract

Pooled CRISPR screens increasingly use longitudinal, donor-adjusted, and factorial designs, whereas beta-binomial screen methods have largely remained limited to two-group comparisons. BARCS extends the library-total-conditional beta-binomial model to guide-level regression with an arbitrary design matrix, enabling direct estimation of time, covariate, and interaction effects. We evaluated BARCS in longitudinal and ordered-phenotype experiments, multivariable simulations with known truth, and external false-discovery audits. In a three-time-point Cas13 analysis, BARCS distinguished proxy-null guide trajectories that competing count methods called significant, although alternative methods ranked essential genes more strongly on the processed matrix. In an IL2RA FACS screen, donor-adjusted analysis of all four ordered bins recovered more validated regulators per discovery than matched MAGeCK-MLE, and held-out non-targeting controls corrected excess guide-level significance. Simulations showed gains from dispersion moderation and control-based denominators under composition shifts. In the external audit, 152 CB^2 null calls fell to zero when full-library probabilities were used; BARCS retained a near-nominal guide-level null in PSN1 but recovered only one of 28 curated DeWeirdt interactions. BARCS therefore generalizes beta-binomial regression from pairwise to multivariable CRISPR designs while preserving full-library denominators; confirmatory use requires independent biological replication and held-out or cross-fitted controls.

Keywords: CRISPR screen; beta-binomial regression; longitudinal design; count data; calibration

1 Introduction

Pooled CRISPR screens increasingly measure guide abundance across time, ordered FACS fractions, donors, doses, or factorial treatment arms. Their scientific targets are slopes, adjusted associations, and interactions. Reducing these experiments to repeated high-versus-low contrasts discards intermediate observations and prevents one effect from being estimated conditional on the remaining design variables.

The count denominator is central to this problem. RNA-sequencing methods generally model marginal counts with negative-binomial distributions and represent library size through normalization or an offset [1–4]. In a pooled screen, however, each guide count is observed together with the total number of mapped guide reads in that library. Conditional on this total, guide abundance is a proportion. The beta-binomial model represents that sampling structure directly and separates depth-dependent sequencing variation from heterogeneity among biological libraries. The original CB^2 study found better goodness of fit for beta-binomial than negative-binomial models in pooled CRISPR counts [5].

CB^2 applies this principle to two independent groups, whereas modern screen designs require a general coefficient model. Beta-binomial regression is available in `corncob` and statistical packages including `aod`, `VGAM`, and `gamlss`

[6–9], but these tools do not provide a CRISPR-oriented workflow that preserves immutable full-library totals, fits guide-level design matrices at screen scale, and connects the result to established screen outputs. Specialized methods such as Chronos and Waterbear model population dynamics or joint FACS-bin membership, respectively, but are tied to those experimental structures [10, 11].

We developed BARCS (Beta-binomial Analysis and Regression for CRISPR Screens) to fill this design gap. BARCS keeps the library-total-conditional beta-binomial sampling model and replaces two group means with a regression design matrix. A single fit can estimate time, donor-adjusted abundance, treatment, or interaction effects. Independent biological libraries, rather than individual reads, determine the residual degrees of freedom. This generalizes the questions available through CB^2; it does not require the two implementations to share the same estimator or finite-sample test.

We evaluate the extension in the order in which it is used. A three-time-point Cas13 screen tests longitudinal coefficients across four replicate-complete cell lines. An IL2RA screen tests a donor-adjusted ordered phenotype against independently validated regulators and held-out non-targeting guides. Genome-scale and factorial simulations then isolate dispersion moderation and denominator choice under known truth. Finally, an external audit shows how pre-fit subsetting can change the effective library total and the apparent false-discovery result. Together, these analyses test the central claim that beta-binomial CRISPR analysis can be extended from pairwise comparisons to interpretable multivariable coefficients while retaining the denominator that defines the sampling model.

2 Results

We first define the specific extension beyond CB^2, then test that extension in longitudinal Cas13 and donor-adjusted IL2RA screens. Simulations isolate the contributions of dispersion moderation and denominator choice, and an external audit examines how preprocessing changes apparent false-discovery behavior. Endpoint, serial-harvest, copy-number, and multi-condition follow-up analyses are reported in the Supplement.

2.1 BARCS generalizes the CB^2 sampling model

CB^2 conditions each guide count on the corresponding full-library total, separating sequencing variation from between-library heterogeneity [5]. BARCS retains that sampling principle but replaces the two-group mean comparison with a regression coefficient. Time, dose, donor, batch, and interaction terms can therefore be estimated in one model rather than reduced to pairwise contrasts. This is the methodological claim tested below: BARCS generalizes the questions that can be asked with the beta-binomial screen model. It does not claim numerical equivalence to the legacy CB^2 statistic, which uses different dispersion, weighting, and degrees-of-freedom calculations.

2.2 Processed Cas13 data provide a longitudinal sensitivity analysis

The Cas13 fitness screens of Liang et al. [12] contain measurements from days 0, 7, and 14. K562 was excluded because one day-0 replicate was not deposited. In HAP1, HEK293FT, MDA-MB-231, and THP1, every method estimated the same day-0-to-day-14 slope with a replicate block and a two-sided test. Thus the comparison preserves the longitudinal feature rather than collapsing the screen to an endpoint.

The principal finding is a calibration–ranking tradeoff. BARCS produced a lower mean cell-line-specific proxy-null calibration error than MAGeCK-MLE (0.025 versus 0.057), but MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom ranked known essential genes more strongly (average precision 0.874–0.877 versus 0.838). At a matched empirical 5% proxy-null false-positive rate, BARCS recalled 0.863 of essential genes and the alternatives recalled 0.879–0.888. BARCS therefore reduced excess significance in this input without improving essential-gene ranking.

Figure 1 localizes that difference. Three guides assigned to the proxy-null set had BARCS p-values of 0.20–0.23

but were significant under DESeq2, edgeR-QL, and limma–voom. Their two replicate trajectories do not show a consistent monotone depletion from day 0 through day 14. These are concrete disagreement cases, not independent biological validation.

The deposited matrix had already been normalized, ComBat-corrected, and outlier-processed before it was rounded to pseudo-counts. The result is therefore a same-input sensitivity analysis, not a likelihood comparison. Raw-read confirmation is required before attributing the observed differences to beta-binomial or negative-binomial sampling assumptions.

2.3 A covariate-adjusted ordered phenotype recovers validated regulators

We next asked whether the design-matrix extension is useful outside a time course. GSE242880 contains an IL2RA protein-abundance screen in primary human T cells, with four ordered FACS bins for each of three donors, 6,000 guides, and 26 regulators validated by individual knockout and flow cytometry [11, 13]. The four bins were assigned prespecified latent normal locations, and BARCS fitted one marker-abundance slope with donor indicators. Official MAGeCK-MLE received the same design.

This experiment also provides a direct model diagnostic. Bins from one donor partition the same cell pool and are negatively dependent, whereas the guide-level BARCS fit treats their margins independently. Among 593 non-targeting guides, the uncalibrated $p < 0.05$ frequency was 0.133. Because `bb_calibrate_controls()` estimates its scale from the control tail, evaluating those same guides after fitting would be circular. We therefore used deterministic five-fold cross-fitting: each guide was evaluated with a scale estimated from the other four folds. The held-out $p < 0.05$ rate was 0.049. The production gene analysis used all controls to estimate one scale; calibration does not change effect estimates or their ranking.

Table 1: Ordered-bin IL2RA sensitivity analysis. Recovery requires a discovery at each method's threshold and the experimentally validated direction. The BARCS NTC rate is five-fold held out; it is not measured on the controls used to estimate that fold's scale. Validation yield is the number of validated regulators recovered divided by all discoveries; it is a descriptive efficiency ratio, not an odds ratio.

Method	Discoveries	Validated	Validation yield	NTC $p < 0.05$
■ BARCS, four bins + controls	49	22/26	22/49 (44.9%)	0.049
■ BARCS, four bins raw	127	23/26	23/127 (18.1%)	0.133
■ BARCS, outer bins	34	18/26	18/34 (52.9%)	0.064
■ MAGeCK-MLE, four bins	72	17/26	17/72 (23.6%)	—
■ MAGeCK test, outer bins	30	18/26	18/30 (60.0%)	—
■ Waterbear, four bins [11]	79	24/26	24/79 (30.4%)	—
■ MAUDE, four bins [14]	406	25/26	25/406 (6.2%)	—

Control-calibrated BARCS recovered 22 of 26 validated regulators with 49 discoveries, compared with 17 of 26 and 72 discoveries for the matched MAGeCK-MLE fit (Table 1). This corresponds to validation yields of 44.9% and 23.6%, respectively. The four-bin BARCS model also recovered four regulators missed by its outer-bin fit, showing what the ordered intermediate bins contributed beyond a binary comparison.

The specialist models remain important comparators. Waterbear recovered 24 validated regulators and MAUDE recovered 25, but with 79 and 406 discoveries. BARCS therefore offered the strongest recovery-per-discovery ratio among the four-bin models shown, while Waterbear remained the more faithful generative model for correlated sorted bins. Negative controls diagnose the independence violation and calibrate one operating point; they do not make the bin margins independent.

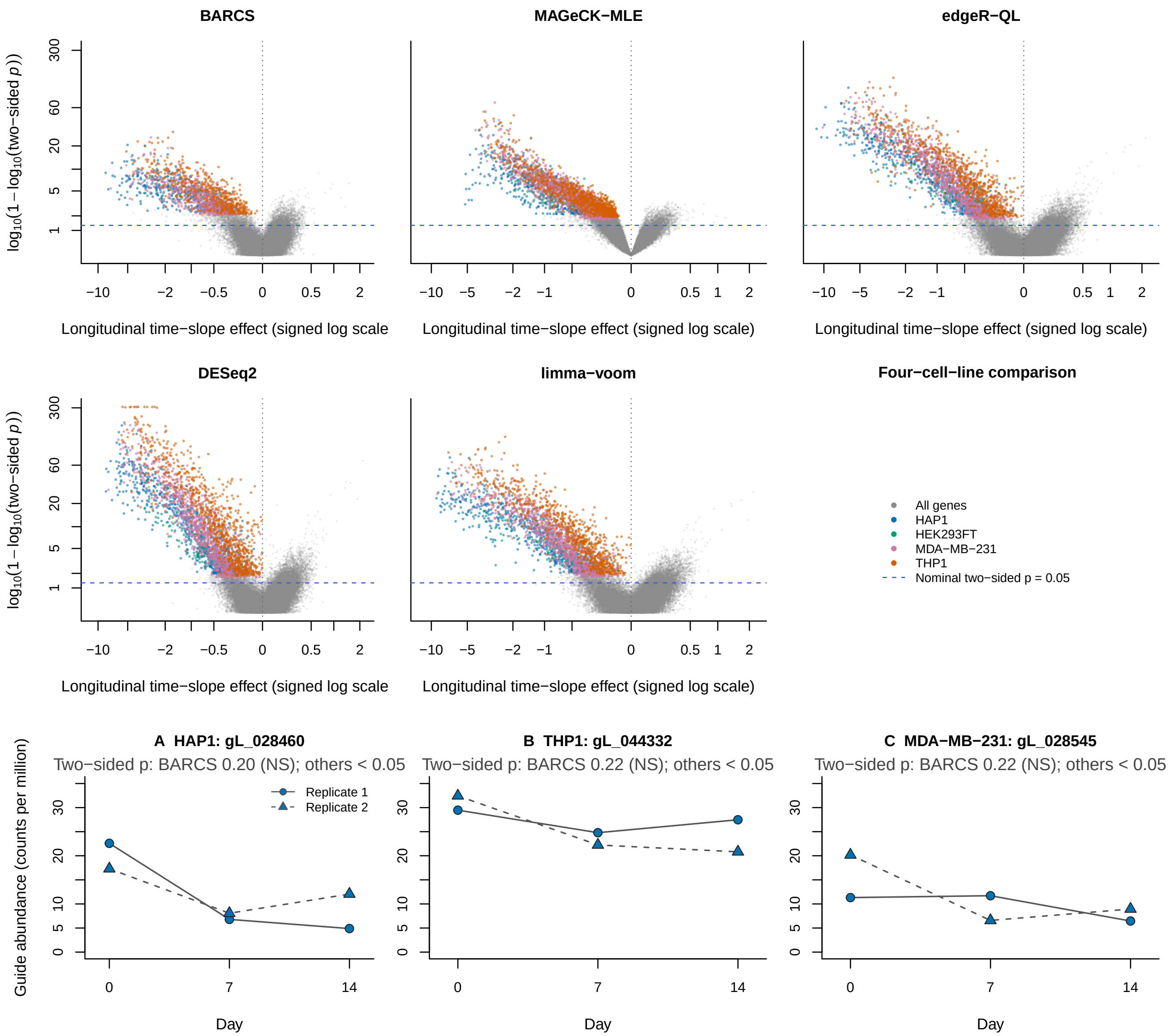

Figure 1: Longitudinal Cas13 sensitivity analysis using days 0, 7, and 14. The upper rows show method-specific volcano plots for BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom, pooling gene-level results from HAP1, HEK293FT, MDA-MB-231, and THP1. K562 is excluded because one day-0 replicate is absent. Within each retained cell line, every method estimates the same continuous time slope from the identical processed-count matrix. Colored points are negatively directed genes called at two-sided FDR 0.10. To make the dense central region legible, the horizontal axis is a signed logarithmic effect transform and the vertical axis is $\log_{10}\{1 - \log_{10}(p)\}$; the dashed line marks two-sided $p = 0.05$. (A–C) Processed guide counts divided by the corresponding library size, in counts per million, for three proxy-null examples. Lines connect measurements from the same replicate across days. BARCS leaves each guide non-significant, whereas DESeq2, edgeR-QL, and limma–voom each call it significant under the proxy-null benchmark. MAGeCK-MLE is included in the gene-level volcano comparison but not in panels A–C because its reported statistic is gene-level.

2.4 Dispersion moderation improves BARCS in genome-scale simulation

We used CRISPulator to evaluate a FACS design under known truth [15]. Three prespecified seeds each generated 10,000 genes, five guides per gene, four independent replicates, and low, bulk, and high fractions. All methods received the same counts and design.

The comparison isolates one change within BARCS. BARCS-ST uses the original guide-specific dispersion estimate, whereas BARCS-MOD moderates those estimates before applying the same gene summary. Moderation increased average precision from 0.902 to 0.919 and F1 at gene FDR 0.10 from 0.811 to 0.847, while realized FDP remained nearly unchanged (0.065 and 0.066). This supports dispersion moderation within the common BARCS pipeline.

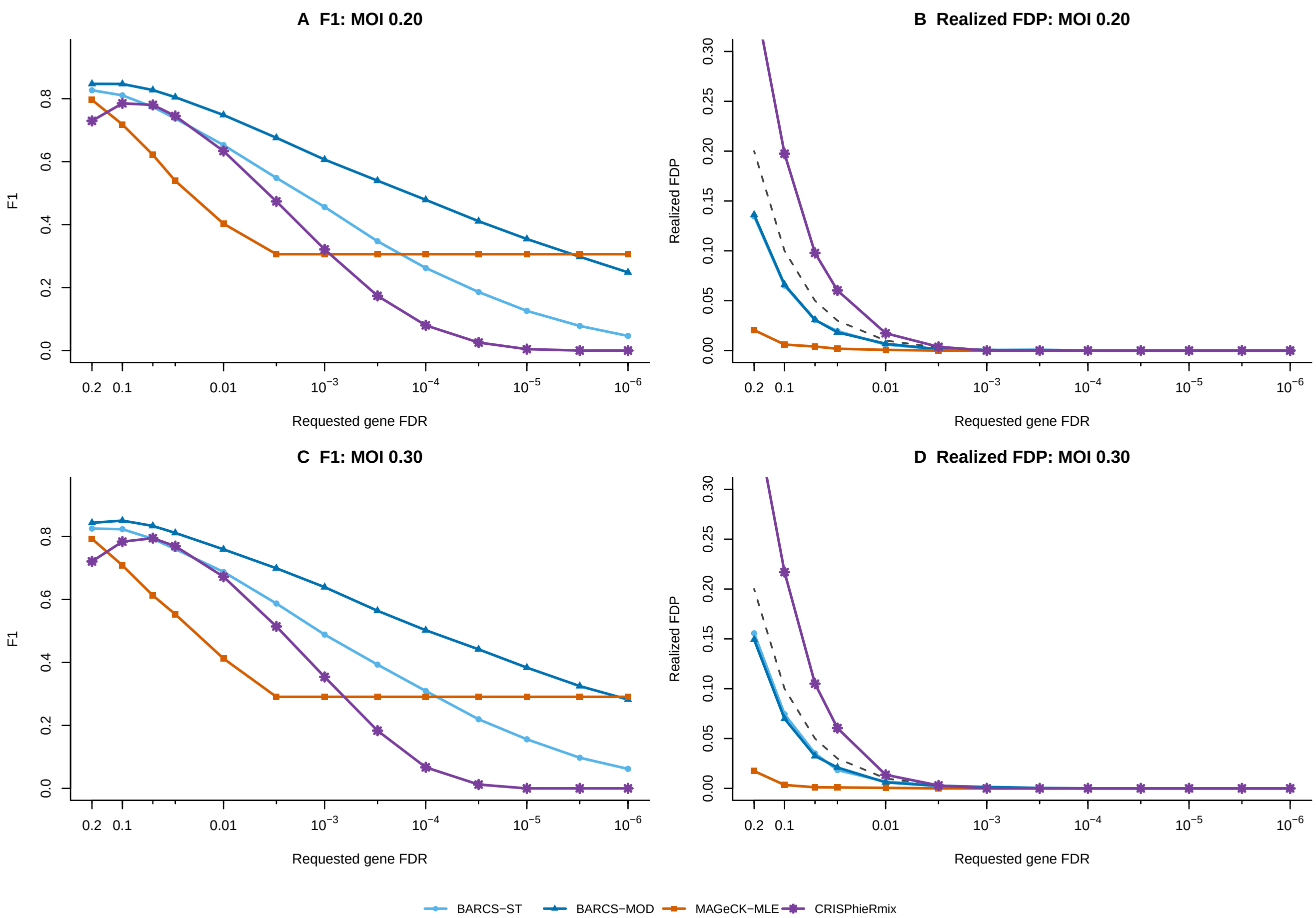

Figure 2: CRISPulator threshold sensitivity. (A,B) MOI 0.20 and (C,D) MOI 0.30. F1 is shown on the left and realized FDP on the right; the dashed diagonal denotes realized FDP equal to the requested threshold. The shared simulation, gene summary, and thresholds isolate the BARCS-ST versus BARCS-MOD moderation effect.

Figure 2 shows that the gain persisted across requested thresholds and at both simulated multiplicities of infection. The genome-scale comparator set was restricted to MAGeCK-MLE and CRISPhieRmix; it therefore supports the moderation ablation, not a universal method ranking. In a broader 400-gene comparison, edgeR-QL, DESeq2, and limma–voom matched or exceeded BARCS ranking but had higher realized FDP. Ranking and calibration thus favored different methods.

2.5 Control denominators recover interactions under composition shifts

We next used simCRISPR to generate a factorial knockout-by-treatment screen with a known guide-level interaction [16]. Each of three seeds contained 2,000 sgRNAs and three independent replicates per factorial arm. Non-targeting guides had a true interaction of zero, and safe-harbor guides separated the effect of cutting from a targeted gene effect. Each control class was split so that normalization and calibration guides were never used for evaluation.

Table 2: Interaction recovery in simCRISPR (three-seed mean). Recall and F1 use targeting guides with |interaction| ≥ 0.2 as positives. Null and safe-harbor rates are held-out call fractions at FDR 0.10.

Method	Denominator	Dir. recall	F1	Null rate	Safe-harbor
BARCS-ST	Library	0.376	0.488	0.004	0.002
BARCS-MOD	Library	0.375	0.481	0.000	0.000
BARCS-ST	Non-targeting	0.769	0.870	0.024	0.070
BARCS-MOD	Non-targeting	**0.838**	**0.917**	0.040	0.080
BARCS-ST	Safe-harbor	0.762	0.864	0.042	0.042
BARCS-MOD	Safe-harbor	0.830	0.914	0.027	**0.035**
MAGeCK-MLE	Non-targeting	0.003	0.006	0.000	0.000

The full-library denominator preserved effect ranking but shifted the true-zero guides because widespread depletion increased every surviving guide's library share. BARCS-MOD consequently reached F1 0.481 with the library denominator, compared with 0.917 and 0.914 using non-targeting and safe-harbor denominators. The held-out null rates remained below the nominal 0.10 in all cases. The result identifies denominator choice, rather than the interaction coefficient itself, as the source of the power loss.

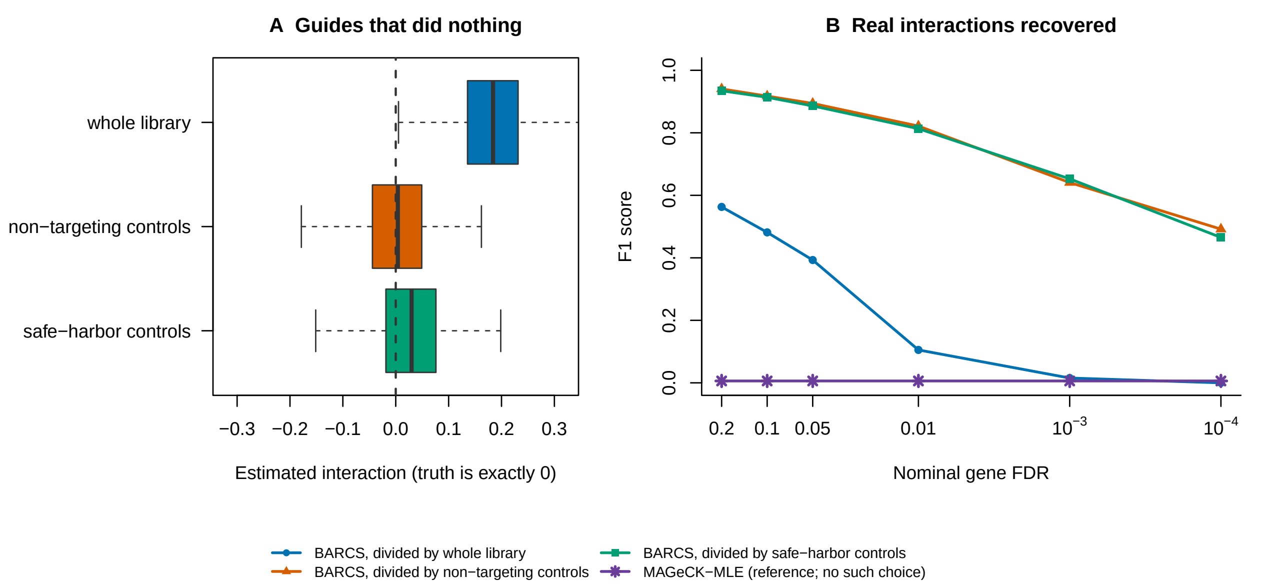

Figure 3: Denominator choice controls interaction recovery. (A) Estimated interactions among true-zero guides. A full-library denominator induces a positive shift; either control denominator removes it. (B) F1 across requested FDR thresholds. Control denominators retain substantially more interaction power.

The two control denominators performed similarly overall but differed on cutting artifacts. Safe-harbor guides were called at 0.080 after non-targeting normalization and 0.035 after safe-harbor normalization. Non-targeting controls are therefore efficient for detecting targeted interactions, whereas safe-harbor controls better absorb the shared cutting response. MAGeCK-MLE is shown only as a reference: the simulated truth is guide-level, leaving no within-gene replication for its native gene model.

2.6 False-discovery conclusions depend on the analysis contract

We audited the deposited benchmark of CB^2, MAGeCK, JACKS, DrugZ, and Chronos [17, 18]. We separated two analyses that answer different questions. For Avana, we re-evaluated the published CB^2 analysis [5] after restoring the full-library denominator; BARCS was not fitted because this audit isolates the denominator used by CB^2. For PSN1 and DeWeirdt, we retained the deposited CB^2 results and ran BARCS on the deposited normalized tables. In the reported Avana analysis, restriction to null genes before fitting changed the column totals. Reusing the full-library CB^2 probabilities for the same genes reduced the number of FDR 0.10 discoveries from 152 to zero.

Table 3: Reported, reanalyzed CB^2, and BARCS results in the external false-discovery audit. Reported results and deposited inputs follow Dempster et al. [17] and the accompanying data release [18]. CB^2 was reanalyzed only for Avana; BARCS was run for PSN1 and DeWeirdt. "Known" denotes the 28 curated DeWeirdt condition–gene combinations [19], not a complete truth set.

Benchmark	Reported by Dempster et al. [17]	CB^2 reanalysis	BARCS	Diagnosis
Avana null-only	152 calls at FDR 0.10	0 calls at FDR 0.10	—	Largest full-total null $p < 0.05$ rate: 0.145
PSN1 null split	0 gene calls	—	0 gene calls; guide $p < 0.05$: 0.0481 raw and 0.0465 scaled	Control scale: 1.017
DeWeirdt differential screens	0 calls; 0/28 known recovered	—	38 calls; 1/28 known interactions recovered	37/38 calls arose in one comparison

The PSN1 result shows that pooling one guide dispersion across the design can avoid the extreme loss of power produced by separate two-replicate variance estimates while retaining a near-nominal guide null. In the DeWeirdt screens [19], however, BARCS recovered only MCL1 under A-1331852; 37 of its 38 calls came from the OVCAR8–olaparib comparison without recovering BRCA1 or BRCA2. A global treatment-associated quality shift cannot be separated from treatment with only two replicates per arm.

These results do not establish a winner against a complete ground truth. They show that pre-fit guide subsetting and common normalization can alter the effective depth seen by a library-total-conditional model. Comparator studies must preserve full-library totals, separate calibration guides from evaluation guides, and use blocked permutations or independent validation for differential screens.

3 Methods

3.1 Model and inference

For guide g in library i, BARCS models the count conditional on the unfiltered mapped-guide total:

$$K_{gi} \mid N_i \sim \mathrm{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \qquad \mathrm{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g. \tag{1}$$

The design vector can contain time, dose, donor, batch, treatment, and interaction terms. Coefficients are estimated by beta-binomial weighted logistic regression. A prespecified coefficient or contrast is divided by its model-based standard error and compared with a Student t distribution using residual sample degrees of freedom. All reported tests are two-sided, and guide- or gene-level probabilities are adjusted by Benjamini–Hochberg. Optional dispersion moderation and control scaling are fitted without changing the coefficient estimate. Full estimation and aggregation details are given in the Supplement.

3.2 Longitudinal Cas13 analysis

For HAP1, HEK293FT, MDA-MB-231, and THP1, days 0, 7, and 14 were fitted jointly with continuous time and a replicate indicator. K562 was excluded because one day-0 replicate was absent. BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom received the identical rounded matrix and tested the same time slope. Known essential genes were positives; unexpressed targeted lncRNAs were proxy nulls. Because the deposited values were already normalized and ComBat-corrected, this analysis was treated as a processed-count sensitivity study.

3.3 Ordered-bin IL2RA analysis

The IL2RA screen comprised four ordered FACS bins from each of three donors. BARCS fitted one ordered-bin score with donor indicators, and MAGeCK-MLE received the same design. Of 593 non-targeting guides, four folds estimated the control scale and the omitted fold evaluated it; this was repeated across five deterministic folds. Method discoveries were compared with 26 regulators validated by individual knockout and flow cytometry. Waterbear and MAUDE results were taken from their published analyses of the same experiment.

3.4 Simulation benchmarks

CRISPulator generated three 10,000-gene FACS screens with five guides per gene and four independent replicates. BARCS-ST and BARCS-MOD differed only by guide-dispersion moderation. simCRISPR generated three factorial knockout-by-treatment screens with 2,000 sgRNAs and three replicates per arm. Its known interaction effects allowed direct comparison of full-library, non-targeting, and safe-harbor denominators. Control guides used for normalization and calibration were held out from scoring.

3.5 External and supplementary analyses

The external false-discovery audit re-evaluated the deposited Avana CB^2 probabilities using full-library totals and ran BARCS on the deposited PSN1 and DeWeirdt normalized tables. Supplementary analyses include the serial-harvest HT-29 screen, a four-coefficient GSE70038 screen with method-specific pathway enrichment, and the Sanson A375 endpoint and copy-number audit. The Supplement reports their complete preprocessing, design, and scoring definitions in the same order as the corresponding results.

4 Discussion and conclusion

BARCS extends a CRISPR-specific beta-binomial sampling principle to arbitrary design matrices. The contribution is not a new probability distribution; it is the ability to estimate time, covariate, and interaction coefficients while conditioning every guide on the full mapped-guide total.

The longitudinal Cas13 analysis demonstrates that extension directly. BARCS used all three time points with a replicate block and returned fewer significant proxy-null trajectories than the alternative count methods. Those alternatives ranked known essential genes more strongly, so the main finding is a calibration–ranking tradeoff on a transformed matrix. Raw reads are needed to determine how much of that difference comes from the sampling model rather than upstream normalization.

The IL2RA experiment provides the clearest real-data efficiency result. Donor-adjusted BARCS recovered 22 validated regulators in 49 discoveries, compared with 17 in 72 for the matched MAGeCK-MLE design. The intermediate FACS bins contributed four regulators beyond the BARCS outer-bin analysis, and five-fold control cross-fitting reduced the held-out non-targeting rate to the nominal level. Waterbear remains better aligned with the joint-bin data structure, but BARCS provides a simple coefficient analysis when ordered phenotypes and covariates must be handled together.

The simulations identify two practical components of that analysis. Dispersion moderation improved precision and F1 without increasing realized FDP in the genome-scale FACS screen. In the factorial screen, control-based denominators prevented global depletion from shifting true-zero interactions and nearly doubled F1 relative to the full-library denominator. The latter result is not a universal recommendation to replace full-library totals: control denominators are appropriate when stable controls define the intended composition reference, and their choice must match whether cutting artifacts should be retained or absorbed.

Finally, the external audit shows why the analysis contract must be explicit. Subsetting guides before fitting changed the beta-binomial denominator and converted 152 reported null calls to zero when full-library totals were preserved. This does not resolve every calibration question, but it demonstrates that equal preprocessing is not necessarily equal statistical input for models that use sequencing depth differently.

The resulting story is consistent across datasets. BARCS generalizes the beta-binomial screen model beyond two groups; longitudinal and ordered-bin examples show what the additional coefficients recover; and simulations and audits identify the dispersion, denominator, replication, and calibration conditions needed for reliable use.

Two conditions delimit confirmatory use. First, the coefficient reference distribution requires independent biological libraries. FACS bins from one donor form a compositional partition, and the HT-29 harvests derive from one passaged culture; these settings require a joint-bin, blocked, or repeated-measures model. Second, calibration must be evaluated independently of calibration fitting. A control used to estimate the scale cannot also provide an unbiased assessment of that scale, so held-out controls, cross-fitting, or an independent experiment are required.

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A Supplementary information

Detailed methods. The following sections provide the estimation and benchmark specifications summarized in the main Methods.

A.1 Estimation by feasible IRLS

Let X be the $m \times q$ full-rank design matrix. Given current values of $\boldsymbol{\beta}$ and ρ, define

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad \mu_i = \frac{\exp(\eta_i)}{1+\exp(\eta_i)}, \tag{2}$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1-\mu_i)}, \qquad w_i = \frac{N_i \mu_i (1-\mu_i)}{1+(N_i-1)\rho}. \tag{3}$$

The coefficient update is the weighted least-squares step

$$\widehat{\boldsymbol{\beta}}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \qquad W = \operatorname{diag}(w_1, \dots, w_m). \tag{4}$$

This is weighted least squares on the logistic working response, not on the raw proportions.

For a fixed fitted mean, ρ is estimated from the Pearson equation

$$\sum_{i=1}^{m} \frac{(K_i - N_i \widehat{\mu}_i)^2}{N_i \widehat{\mu}_i (1-\widehat{\mu}_i)\{1+(N_i-1)\widehat{\rho}\}} = m - q. \tag{5}$$

If the left side at $\rho = 0$ is no larger than $m - q$, the estimate is truncated at zero rather than claiming negative overdispersion. Equations (3)–(5) are alternated to convergence.

The weight has an important limiting behavior:

$$\lim_{N_i \to \infty} w_i = \frac{\mu_i(1-\mu_i)}{\rho} \quad \text{when } \rho > 0. \tag{6}$$

Therefore a deeply sequenced sample cannot acquire unlimited leverage. Once sequencing uncertainty is small, biological heterogeneity sets the information ceiling.

A.2 t-based inference

At convergence, the model-based covariance estimator is

$$\widehat{\operatorname{Cov}}(\widehat{\boldsymbol{\beta}}) = (X^\top \widehat{W} X)^{-1}. \tag{7}$$

The Pearson equation scales the residual variation to approximately one. If the dispersion estimate reaches its numerical upper boundary, the implementation additionally multiplies Equation (7) by the remaining Pearson scale.

Equation (7) is a plug-in, model-based covariance: it conditions on the estimated $\widehat{\rho}$ as though it were fixed. It does not propagate uncertainty from estimating a separate dispersion for every guide. Referring the coefficient to t_{m-q} gives a heavier-tailed small-sample reference than a normal approximation, but it is not a formal correction for dispersion-estimation uncertainty. With few independent libraries, calibration must therefore be checked by simulation, phenotype permutation, or prespecified negative controls.

For coefficient β_j, the statistic

$$t_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\operatorname{Cov}}(\widehat{\boldsymbol{\beta}})_{jj}}} \tag{8}$$

is compared with a Student t_{m-q} distribution. The degrees of freedom are driven by the number of independent sequenced samples, not by the total read count. This is the crucial reason that a library with millions of reads does not create artificial certainty.

More generally, for a prespecified contrast vector $\mathbf{c}$,

$$t_{\mathbf{c}} = \frac{\mathbf{c}^\top \widehat{\boldsymbol{\beta}} - \delta_0}{\sqrt{\mathbf{c}^\top \widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}})\mathbf{c}}} \tag{9}$$

uses the same $m - q$ degrees of freedom. This covers pairwise group contrasts, average slopes in interaction models, and other one-degree-of-freedom hypotheses. Guide-wise two-sided p-values are adjusted using the Benjamini–Hochberg procedure [20].

A.2.1 Optional guide-dispersion moderation

BARCS-MOD is an optional empirical-Bayes transformation of the fitted guide statistics, not a separate likelihood. Let $\nu = m - q$ and let $s_g^2 = Q_g^{(0)}/\nu$ be guide g's Pearson variance inflation under the binomial working model. A lowess trend in log mean CPM gives a guide-specific prior scale s_{0g}^2, and the scaled-F log-moment estimator of Smyth [21] supplies prior degrees of freedom d_0. The moderated inflation is

$$\widetilde{s}_g^2 = \frac{\nu s_g^2 + d_0 s_{0g}^2}{\nu + d_0}. \tag{10}$$

Because the original beta-binomial fit cannot estimate negative overdispersion, its stored standard error corresponds to the fitted inflation $s_{g,\mathrm{fit}}^2 = \max(1, s_g^2)$. The implementation therefore reports

$$\mathrm{SE}_{g,\mathrm{mod}} = \mathrm{SE}_g \sqrt{\frac{\widetilde{s}_g^2}{\max(1, s_g^2)}}, \qquad t_{g,\mathrm{mod}} = \frac{\widehat{\beta}_g}{\mathrm{SE}_{g,\mathrm{mod}}}, \qquad \nu_{\mathrm{mod}} = \nu + d_0. \tag{11}$$

The one-sided `pmax` truncation in Equation (11) is part of the implementation and is not an estimated scientific constraint. The optional `one_way=TRUE` variant additionally replaces $\widetilde{s}_g^2$ by $\max(\widetilde{s}_g^2, s_{g,\mathrm{fit}}^2)$, so moderation can only increase a standard error. All reported BARCS-MOD results are simulation ablations; the real-data benchmarks use unmoderated BARCS.

A.2.2 From guide statistics to gene statistics

BARCS fits the count model guide by guide. The FACS, simulation, GSE70038, A375, and descriptive HT-29 benchmark scripts then use an explicit directional Stouffer summary, denoted BARCS-ST below, to obtain one gene-level statistic. For the jth valid guide targeting gene g, the two-sided guide p-value is converted back to a signed standard-normal score,

$$z_{gj} = \mathrm{sign}(\widehat{\beta}_{gj})\Phi^{-1}\left(1 - \frac{p_{gj}}{2}\right). \tag{12}$$

If m_g valid guides target the gene, their combined score and two-sided gene-level p-value are

$$Z_g = \frac{1}{\sqrt{m_g}} \sum_{j=1}^{m_g} z_{gj}, \qquad p_g = 2\Phi(-|Z_g|). \tag{13}$$

The reported gene effect is the median of the guide coefficient estimates,

$$\widehat{\beta}_g = \mathrm{median}_{j=1,\dots,m_g} \widehat{\beta}_{gj}, \tag{14}$$

and gene-level false-discovery rates are obtained by applying Benjamini–Hochberg to the collection of p_g values. Consequently, concordant guide directions reinforce one another, discordant directions cancel, and one extreme guide has limited influence on the reported effect.

This aggregation is deliberately transparent, but it is not a native hierarchical gene model. Equations (12)–(13) give equal weight to valid guides and use the independence reference $\sqrt{m_g}$; shared sequence artifacts or other correlation can therefore make the gene statistic anti-conservative. A production extension should estimate guide reliability and within-gene dependence or use a hierarchical effect model. In the biological head-to-head benchmarks, this Stouffer–median procedure is used only for BARCS (and for the deliberately naive binomial comparator). MAGeCK, Chronos, BAGEL2, Waterbear, and MAUDE retain their native or deposited gene-level summaries. The exploratory DeWeirdt false-discovery audit used Fisher aggregation as stated in its Results description and is not pooled with these primary benchmark statistics.

Exchangeable normal guide-beta statistic. BARCS-NORM implements a different and deliberately simple model. It does not transform guide p-values and does not use guide-specific standard errors. Instead, for the m_g valid guides targeting gene g, it assumes

$$\widehat{\beta}_{gj} \overset{\text{iid}}{\sim} N(\mu_g, \sigma_g^2). \tag{15}$$

The gene effect and between-guide standard deviation are estimated directly:

$$\widehat{\mu}_g = \frac{1}{m_g} \sum_{j=1}^{m_g} \widehat{\beta}_{gj}, \tag{16}$$

$$s_g^2 = \frac{1}{m_g - 1} \sum_{j=1}^{m_g} (\widehat{\beta}_{gj} - \widehat{\mu}_g)^2. \tag{17}$$

Testing $H_0 : \mu_g = 0$ gives

$$T_g = \frac{\widehat{\mu}_g}{s_g/\sqrt{m_g}} \sim t_{m_g-1} \quad \text{under Equation (15) and } H_0. \tag{18}$$

Thus normally distributed guide betas lead to a Student t reference when their unknown σ_g is estimated from the same guides. The software also reports $2\Phi(-|T_g|)$ as a plug-in normal sensitivity value, but it is not the primary p-value because it treats the estimated spread as known. Multiple guides support this guide-consistency calculation; they do not become additional biological screen replicates.

Random-effects summaries. BARCS-RE (the implementation label `BARCS-partial`) retains each guide-specific regression standard error and models

$$\widehat{\beta}_{gj} \mid \beta_g, \tau_g^2 \sim N\Big(\beta_g,\, \mathrm{SE}(\widehat{\beta}_{gj})^2 + \tau_g^2\Big). \tag{19}$$

It estimates τ_g^2 by the nonnegative DerSimonian–Laird estimator [22] and then pools guide effects with weights

$$w_{gj} = \Big\{\mathrm{SE}(\widehat{\beta}_{gj})^2 + \widehat{\tau}_g^2\Big\}^{-1}, \qquad \widehat{\beta}_g = \frac{\sum_j w_{gj}\widehat{\beta}_{gj}}{\sum_j w_{gj}}. \tag{20}$$

BARCS-RE-EB (implementation label `BARCS-EB`) stabilizes the noisy gene-specific heterogeneity estimate by shrinking it toward the pooled screen-wide excess-Q estimate τ_0^2:

$$\widetilde{\tau}_g^2 = \frac{(m_g - 1)\widehat{\tau}_g^2 + d_0\tau_0^2}{(m_g - 1) + d_0}, \qquad d_0 = 4. \tag{21}$$

Both random-effects statistics are calibrated against prespecified negative-control genes when at least ten are available and otherwise against a robust whole-screen empirical null. In contrast, BARCS-NORM obtains its spread and reference degrees of freedom entirely within each gene. These three alternatives and BARCS-ST reuse exactly the same fitted guide coefficients; only the guide-to-gene inference changes.

Guide-consistency statistic for the one-replicate boundary. When sample replication is insufficient, converting a guide statistic through a low-degree-of-freedom t distribution can erase useful ranking information. We therefore prespecified a separate exploratory gene statistic, BARCS-GC, for the $R = 1$ CRISPulator analysis. It uses the uncalibrated guide coefficient and model-based standard error to estimate one shared gene effect. With

$$w_{gj} = \mathrm{SE}(\widehat{\beta}_{gj})^{-2}, \tag{22}$$

the inverse-variance estimate, its standard error, and raw Wald statistic are

$$\widehat{\beta}_g = \frac{\sum_{j=1}^{m_g} w_{gj}\widehat{\beta}_{gj}}{\sum_{j=1}^{m_g} w_{gj}}, \qquad \mathrm{SE}(\widehat{\beta}_g) = \left(\sum_{j=1}^{m_g} w_{gj}\right)^{-1/2}, \qquad T_g = \frac{\widehat{\beta}_g}{\mathrm{SE}(\widehat{\beta}_g)}. \tag{23}$$

This is a direct fixed-effect estimate of the common guide coefficient, not a Fisher or Stouffer combination of guide p-values. Its additional assumption is that the guide coefficients target one common gene effect; the reported direction-agreement fraction exposes gross heterogeneity.

Because T_g does not have a trustworthy sample-replicate reference distribution at $R = 1$, it is calibrated across genes. Let c_0 be the all-gene median and

$$s_{\mathrm{MAD}} = \frac{\mathrm{median}_g\, |T_g - c_0|}{\Phi^{-1}(0.75)}. \tag{24}$$

When at least ten valid control genes are present, let c_C be their median and let

$$s_C = \frac{Q_{0.95}\{|T_g - c_C| : g \in C\}}{\Phi^{-1}(0.975)}. \tag{25}$$

The calibrated statistic and two-sided empirical-null p-value are

$$Z_g^{\mathrm{GC}} = \frac{T_g - c_C}{\max(1, s_{\mathrm{MAD}}, s_C)}, \qquad p_g^{\mathrm{GC}} = 2\Phi(-|Z_g^{\mathrm{GC}}|), \tag{26}$$

followed by Benjamini–Hochberg adjustment. If too few control genes are available, c_0 replaces c_C and s_C is omitted. At least three finite guide coefficient estimates are required per gene. The all-gene MAD assumes that fewer than half of tested genes are active.

BARCS-GC measures reproducibility across independently designed perturbations; it does not turn guides into independent biological samples. Its p-value is consequently labeled empirical-null guide-consistency evidence, not sample-level confirmation. The ordinary BARCS result remains the primary analysis when biological replicates are available, and independent experimental replication remains necessary for confirmatory claims.

A.2.3 Liang Cas13 longitudinal fitness benchmark

We analyzed the transcriptome-scale RfxCas13d fitness screens of Liang et al. [12] in HAP1, HEK293FT, MDA-MB-231, and THP1 cells. The library contains 56,322 guides targeting 5,496 lncRNAs, 3,156 protein-coding genes, and 1,000 non-targeting controls. Deposited Table S2 contains guide measurements at days 0, 7, and 14. The previous endpoint analysis selected only days 0 and 14; the present analysis uses all three time points.

The deposited values are fractional after median-of-ratios normalization, ComBat correction, and replicate-outlier processing. They are therefore not literal sequencing counts. BARCS requires integer observations, so each value was rounded once to the nearest pseudo-count and the identical rounded matrix was supplied to every newly fitted method. The mean absolute rounding change was approximately 0.25 and the maximum was less than 0.5 in every cell line. Consequently, this benchmark is a same-input sensitivity analysis rather than a likelihood-faithful raw-count comparison.

BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom fitted a linear longitudinal effect with time scaled from 0 to 1 over 14 days and a replicate indicator. K562 was excluded because one deposited day-0 replicate is absent, rather than fitting an unmatched trajectory without the replicate block. BARCS guide statistics were calibrated with the non-targeting controls and combined with the prespecified directional Stouffer summary. BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom used two-sided guide or Wald probabilities. For the four guide-level regression methods, coefficient direction was retained in the signed Stouffer score; the resulting gene probability was two-sided and was adjusted by the Benjamini–Hochberg procedure. Endpoint RRA summaries were excluded because they do not estimate the three-time-point longitudinal coefficient.

Known-essential protein-coding genes were positives. Cell-line-specific nulls were targeted lncRNAs with TPM equal to zero in both available expression assays. We reported average precision, essential-control recall at an empirical 5% null false-positive rate, the absolute deviation of the null $p < 0.05$ fraction from 0.05, and directional essential-control recall at gene FDR 0.10. A restartable raw-read pipeline is also provided for BioProject PRJNA1344834 using the stated Cutadapt anchors [23] and Bowtie `-v 1 -m 3 -best -q` [24].

A.3 Ordered-bin IL2RA benchmark

The GSE242880 count matrix contains four ordered FACS bins for each of three donors, 6,000 guides, and 593 non-targeting guides [11, 13]. Bin locations were set to the expected values of four equal-probability intervals under a standard normal distribution. BARCS fitted this ordered score with donor indicators. MAGeCK-MLE received the same design matrix; the outer-bin MAGeCK test used only the low and high fractions.

BARCS guide results were aggregated with the prespecified directional Stouffer summary. For held-out calibration, non-targeting guides were ordered by guide identifier and assigned cyclically to five folds. Each fold was evaluated with the tail scale estimated from the other four folds. The production gene analysis used all controls, whereas the reported non-targeting $p < 0.05$ rate uses only held-out guide predictions.

Directional recovery was evaluated against 26 regulators validated by individual knockout and flow cytometry. A recovered regulator had to pass the method's reported threshold and agree with the validated direction. Validation yield was calculated as recovered regulators divided by all discoveries. Published Waterbear and MAUDE totals were used only for this recovery comparison; no unavailable non-targeting rate was imputed.

A.3.1 CRISPulator FACS benchmark

We used CRISPulator 0.5.1's documented interfaces for library construction, transfection, FACS selection, and sequencing [15]. The committed Julia environment pins the tagged source revision. For each simulation seed, one CRISPRn guide library was constructed and reused across four independently seeded transfection, sorting, and sequencing replicates. The default CRISPulator phenotype mixture was retained: 75% inactive, 5% negative-control, 10% phenotype-increasing, and 10% phenotype-decreasing genes. The headline screens contained 10,000 genes and five guides per gene, so 50,000 guides, which is the order of a genome-wide library rather than a pilot. The default guide-quality mixture—90% complete-knockout guides and 10% low-activity guides—was retained. Transfection representation was 500 cells per guide, Gaussian phenotype noise was 2, sorting representation was 50 cells per guide, and sequencing depth was 50 reads per guide per sample. Multiplicity of infection was 0.20 for the main analysis and 0.30 for the supplementary analysis; 0.20 is the point at which the single-integration approximation these guide-level models share is most defensible, and 0.30 stresses it. Both multiplicity of infection and the high-quality-guide fraction are exposed as simulation parameters, as is the number of genes and the number of independent screen replicates. The three genome-scale benchmark seeds were 20250724 through 20250726; the supporting 400-gene analyses below use 20250724 through 20250728.

Two BARCS gene statistics are carried through the genome-scale ablation. BARCS-ST is the historical calibrated signed-z statistic. BARCS-MOD reuses the identical guide fits after guide-dispersion moderation, so the two differ only in how the guide variance is estimated, not in the gene combiner. The exchangeable-normal, random-effects, and moderated random-effects statistics are retained for the real-data comparisons but are not carried through the genome-scale simulation, where the question is whether moderation changes the calibration–power tradeoff rather than which of several pooling rules is best. Official MAGeCK-MLE 0.5.9.5 and CRISPhieRmix are included in that run, but this restricted set is not treated as an exhaustive comparator panel. MAGeCK-MLE's ordered marker is affinely mapped onto zero–one so that the low samples of the reference replicate form the all-zero reference row the MAGeCK initializer requires. Because MAGeCK-MLE omits negative-control genes from its gene summary once their sgRNAs are declared, all methods in the genome-scale benchmark are scored on the genes every method returned a finite result for, 9,475–9,517 of 10,000 per run. The negative-control diagnostic is instead taken from each method's own output, and is therefore unavailable for MAGeCK-MLE.

We additionally performed a one-factor-at-a-time sensitivity analysis around the baseline $(\mathrm{MOI}, q, G, R) = (0.25, 0.90, 400, 4)$, where q is the high-quality-guide fraction, G is the number of genes, and R is the number of independent screen replicates. The evaluated levels were $\mathrm{MOI} \in \{0.10, 0.25, 0.40\}$, $q \in \{0.60, 0.75, 0.90\}$, $G \in \{100, 400, 1000\}$, and $R \in \{1, 3, 4, 6\}$. After removing duplicate baseline settings, this yielded ten scenarios; each used the same five prespecified seeds, for 50 simulations. One parameter changed at a time, so its main sensitivity remained interpretable. For each seed and scenario, we paired BARCS and MAGeCK-MLE results from the low–bulk–high design and reported differences as BARCS minus MAGeCK-MLE. The executable benchmark also supports the complete $3 \times 3 \times 3 \times 4$ factorial (108 scenarios and 540 simulations across five seeds) when interaction effects are the target.

The $R = 1$ setting was prespecified as a diagnostic boundary rather than a confirmatory design. Its three low–bulk–high samples and two regression coefficients leave only one residual degree of freedom. The corresponding two-tail fit has two samples and two coefficients, leaves zero residual degrees of freedom, and was therefore not fitted. All pairwise-design comparisons and the primary performance interpretation use $R \geq 3$; the one-replicate result is shown separately to expose the consequence of insufficient replication. BARCS-GC was evaluated only at $R = 1$ using Equations (23)–(26); it was not used to replace ordinary BARCS in supported replicated designs.

CRISPulator's arbitrary percentile ranges were set to $(0, 0.25)$ for low, $(0.75, 1)$ for high, and $(0, 1)$ for bulk. Thus bulk deliberately overlaps the tails and is not a third component of a multinomial partition. An input sample was generated as an independent sequencing draw from the same guide frequency vector as bulk. Low and high received phenotype scores equal to the means of their corresponding standard-normal intervals,

$$d_{\mathrm{low}} = -1.271, \qquad d_{\mathrm{high}} = 1.271,$$

while bulk received $d_{\mathrm{bulk}} = 0$. BARCS and MAGeCK-MLE fitted this score with replicate indicators [25]. Tail-only fits removed bulk without changing the low and high counts. BARCS guide tests were scaled with the simulated negative-control guides before directional Stouffer aggregation; MAGeCK-MLE used its native gene-level Wald result. The separate 400-gene, five-seed baseline includes edgeR-QL, DESeq2, and limma–voom in addition to BARCS and MAGeCK-MLE. It is used to prevent the restricted genome-scale set from supporting a general method-ranking claim. CRISPhieRmix [26] was run on guide-level $\log_2$ fold changes from a DESeq2 Wald fit of the same design [2], which is the input its documentation specifies; DESeq2 therefore appears as a preprocessing step in the 10,000-gene run, while its native result is scored in the separate 400-gene comparison. CRISPhieRmix was run with its bimodal alternative, because this screen contains both phenotype-increasing and phenotype-decreasing genes and the one-sided default would score only one direction. Its gene effect is the mean guide $\log_2$ fold change and its local false-discovery rate stands in for a p-value wherever a ranking is required, since the model reports neither a p-value nor a signed gene effect. Reported runtimes cover the primary low–bulk–high model fit and exclude simulation, file input/output, and guide-to-gene aggregation.

Genes from the increasing and decreasing classes were positives. Ranking used the two-sided gene significance score. AUROC and average precision measure active-gene discrimination. Effect recovery is Spearman correlation between the inferred gene effect and CRISPulator's median theoretical guide phenotype among active genes. Directional recall is the fraction of active genes with gene FDR below 0.10 and an effect sign matching the simulated class. Realized FDP is the fraction of called genes that were inactive, and F1 uses active versus inactive status at the same FDR threshold.

A.3.2 simCRISPR interaction benchmark

CRISPulator assigns each guide a single effect and sorts cells on a phenotype, so it cannot generate the design BARCS is aimed at. simCRISPR [16] simulates a factorial screen instead: knockout induction crossed with treatment, propagated over several days of cell growth, with PCR amplification and sequencing applied afterwards. Every sgRNA therefore carries a knockout effect, a treatment effect, and an interaction between the two, and that interaction is a coefficient in a design matrix rather than a contrast of two arms.

Each of three seeds generated 2,000 sgRNAs, of which 300 were non-targeting and 200 were safe-harbor controls, in three independent replicates of each of the four arms, so twelve libraries and eight residual degrees of freedom. Growth was logistic. The alternative exponential model is unbounded: over five days a single guide reached a fifth of the library and library sizes spanned a 175-fold range, which does not represent a pooled screen. Initial library columns were dropped, since they are not part of the factorial contrast. Amplification used the package defaults and each library was sequenced to 500 reads per guide. BARCS fitted $\mathrm{logit}(\mu_{gi}) = \beta_0 + \beta_1 k_i + \beta_2 t_i + \beta_3 k_i t_i$ for knockout indicator k_i and treatment indicator t_i, and tested β_3.

Three denominators were compared. The library denominator is the full column total, BARCS's default. The two control denominators hold a chosen control class at a constant share, rescaled to the mean library size and rounded, so that they remain interpretable as library sizes. The two control classes are not interchangeable: safe-harbor guides are cut and carry the DNA-damage response, non-targeting guides are neither. Each control class was split in half, one half normalizing and calibrating and the other held out, so that no guide used to fit the null was also scored against it. MAGeCK-MLE received the same design matrix with an explicit interaction column, the same calibration-half controls, and one sgRNA per gene, because the simulated truth is guide-level.

CRISPhieRmix was not run on these data. It borrows strength across the guides within a gene, whereas simCRISPR assigns every sgRNA an independent interaction, so grouping guides into synthetic genes would impose the structure that the model exploits.

Every targeting guide receives a nonzero interaction, most far too small to resolve, so a magnitude was fixed before scoring: guides with $|\mathrm{INT}| \geq 0.2$ are the positives and the held-out non-targeting guides, whose interaction is exactly zero, are the negatives. Targeting guides below that magnitude are scored for effect recovery but excluded from the detection metrics. Effect recovery is the Spearman correlation between the estimated interaction coefficient and the simulated one across all targeting guides, which needs no threshold. The held-out control call rate and the safe-harbor call rate are reported separately: the first measures false positives against a ground-truth zero, the second measures how often the cutting response alone is mistaken for an interaction.

A.3.3 Longitudinal HT-29 benchmark

Guide counts for the HT-29 time course of Tzelepis et al. [27] were obtained from Data S2 of that article through the Europe PMC supplementary bundle for PMC5081405. The three sequencing columns reported at each of days 7, 10, 13, 16, 19, 22, and 25 were summed to one column per harvest and combined with the pDNA measurement, and guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes. Library totals were computed before that filter and never recomputed, so the beta-binomial denominator remains the full mapped-read total for each sample; the filter removes 4.2% of guides but only 0.10% of reads, uniformly across samples. BARCS and official MAGeCK-MLE 0.5.9.5 were both fitted to this eight-column matrix under an identical design, in which pDNA is day 0 and the tested predictor is day divided by 25, so the two continuous-time fits differ only in the likelihood. A minimum total count of 30 was applied within the BARCS fit. Because all late samples are serial harvests of one culture, these fits are used as descriptive trajectory scores; their residual-t or Wald probabilities are not interpreted as independent-sample inference.

Comparator effects were not recomputed. The joint Chronos, per-endpoint MAGeCK, and per-endpoint BAGEL2 effect matrices were taken as deposited in Chronos Figshare article 14067047 [28], as were the reference essential and nonessential gene lists; the corresponding individual file identifiers are 26548532, 26548541, 30847513, 26548550, and 26548553. The latter two matrices carry one column per day; the day-25 column is used directly, so the comparator is a single observed endpoint at the deepest time point rather than a summary across days.

Two external annotations define the evaluation sets. Positives are the deposited reference essential genes. Negatives are HT-29 genes with DepMap Public 20Q2 expression below 0.5, taken from the ACH-000552 column of Figshare file 26548487, which is the expression source used for the same threshold in the deposited Chronos analysis. The copy-number audit uses the ACH-000552 column of Figshare file 26548484 [29], and correction was performed with MAGeCK 0.5.9.5's official piecewise `-cnv-norm` normalizer applied after model fitting. Because that profile was measured independently of the screen, cell-line drift is a limitation of the copy-number analysis.

The deposited workflow records the source identifier and checksum of every downloaded input, executes the official MAGeCK model and copy-number normalizer, and reproduces the tables reported here.

Additional analyses. These analyses extend the main evaluation without changing its primary longitudinal and multivariable claims.

A.4 HT-29 serial harvests are descriptive, not replicated longitudinal inference

The HT-29 dataset of Tzelepis et al. [27] contains one plasmid pool and guide counts from days 7, 10, 13, 16, 19, 22, and 25 of one passaged culture. The three sequencing columns at each late day were summed because they assay the same harvested population. The seven harvests are likewise repeated measurements of one biological lineage, not seven independent biological units. Referring a time coefficient to t_6 would therefore be pseudoreplication and would violate the independent-library assumption.

We retain the fitted BARCS and official MAGeCK-MLE time coefficients only as descriptive trajectory scores. They received the same eight-column matrix and numeric design. We also report the deposited all-time-point Chronos score and day-25 MAGeCK and BAGEL2 scores [10, 28, 30]. Core essential genes are positives and HT-29 genes with DepMap 20Q2 expression below 0.5 are negatives, matching the deposited evaluation.

Supplementary Table S1: Descriptive HT-29 control-gene ranking. The BARCS and MAGeCK time rows treat serial harvests as a numeric trajectory for ranking only; their sample-level p-values are not interpreted. Recall_{90} is maximum recall at at least 90% precision, and $\text{FP}_{10\%}$ counts unexpressed genes in the lowest 10% of all evaluated effects.

Method	AUROC	PR AUC	NNMD	Recall_{90}	$\text{FP}_{10\%}$
■ BARCS time	0.9786	0.9494	−16.34	0.9037	7
■ Official MAGeCK time	0.9789	0.9534	−17.75	0.8972	5
■ Chronos joint	0.9747	0.9356	−27.88	0.8991	10
■ Deposited MAGeCK day 25	0.9792	0.9428	−14.83	0.9120	10
Deposited BAGEL2 day 25	0.9698	0.9388	−12.06	0.8630	4

The day-25 MAGeCK row has the highest AUROC and Recall_{90} in the deposited comparison. The full trajectory therefore provides no measured advantage over the endpoint on these two ranking metrics. Official continuous-time MAGeCK has the highest PR AUC, Chronos has the most negative NNMD, and BAGEL2 has the fewest unexpressed genes in the 10% tail. These differences illustrate that the metrics reward different score properties; none supplies the missing biological replication.

How a deposited time course is summarized also matters. Taking the median MAGeCK effect across all seven days instead of the day-25 column lowers AUROC from 0.9792 to 0.9742 and Recall_{90} from 0.9120 to 0.8602. Early harvests carry little depletion, so an across-day median attenuates the endpoint signal. This data-reduction effect is larger than the differences among the trajectory scores and further argues against treating this dataset as inferential evidence for one count model.

A.5 Copy-number sensitivity in the longitudinal HT-29 screen

We extracted the HT-29 profile (ACH-000552) from DepMap Public 20Q2 [29] and repeated the same copy-number audit used for A375. Before correction, effect–copy-number Spearman correlation among unexpressed genes is already small: −0.072 for BARCS, −0.075 for official continuous-time MAGeCK, −0.048 for deposited Chronos joint effects, −0.057 for the published day-25 MAGeCK effects, and +0.029 for the published day-25 BAGEL2 effects. Applying MAGeCK 0.5.9.5's official piecewise normalizer makes the association larger in absolute value rather than smaller: −0.145 for BARCS and −0.149 for MAGeCK. Thus the improvement observed in A375 does not generalize automatically.

The piecewise correction estimates one global effect–CNV trend across genes in one cell line and applies it after model fitting. When the nuisance association is weak and biological dependencies are not exchangeable across copy-number regions, the fitted trend can over-correct null genes. Chronos joint is shown without CNV correction because the published procedure requires multiple cell lines to identify common essential genes and was not applied to this single-line experiment [10]. The DepMap profile was measured separately from the original screen, so cell-line drift is an additional limitation. These results support diagnosing copy-number correction rather than applying it by default.

A.6 Complementary hits in a four-condition screen

GSE70038 contains 64,747 guides in 16 libraries from two glioblastoma stem-cell lines and two neural stem-cell lines [31]. BARCS and MAGeCK-MLE received the same five-column design: one shared initial condition and four cell-line-specific terminal coefficients. This analysis tests whether the multivariable interface estimates all four effects jointly; MAGeCK agreement is not treated as biological ground truth.

To examine the disagreements directly, we selected the 200 most depleted genes from each method for each terminal coefficient. This equal-size rule avoids comparing differently calibrated FDR cutoffs. The lists shared 139–144 genes, leaving 56–61 BARCS-only and the same number of MAGeCK-only genes per cell line (Supplementary Table S2).

Supplementary Table S2: Complementary GSE70038 top-200 depletion sets and their strongest Reactome 2022 enrichment. Values in parentheses are Enrichr-adjusted p-values.

Coefficient	Shared	BARCS only	MAGeCK only	Top BARCS-only pathway	Top MAGeCK-only pathway
GSC0131	142	58	58	Cell cycle (1.35×10^{-6})	rRNA processing (1.65×10^{-5})
GSC0827	139	61	61	RNA metabolism (2.77×10^{-9})	RNA metabolism (2.35×10^{-4})
NSCCB660	140	60	60	RNA metabolism (2.10×10^{-4})	RNA metabolism (2.21×10^{-6})
NSCU5	144	56	56	RNA metabolism (3.36×10^{-12})	RNA metabolism (2.80×10^{-13})

Enrichr analysis of the complementary sets used GO Biological Process 2023 and Reactome 2022 [32]. Both methods' exclusive hits formed coherent essential-process groups. BARCS-only GSC0131 hits emphasized cell cycle regulators, whereas the MAGeCK-only set emphasized rRNA processing; RNA metabolism dominated both methods in the other coefficients. These results provide a biological follow-up to the method disagreement, but not an accuracy ranking: Enrichr uses its library background rather than the genes testable in this screen, and essential-process enrichment is expected in a depletion experiment. The complete overlap and enrichment records are provided with the analysis output.

A.7 Endpoint benchmark and copy-number audit

We used the Brunello A375 negative-selection screen of Sanson et al. [33], bundled with CB^2, as a descriptive endpoint benchmark. Reference essential genes are positives and reference nonessential genes are negatives [34]. The common copy-number-complete set contains 1,506 essential and 869 nonessential genes, so its positive prevalence is 63.4%. F1 and recall in this enriched set must therefore be interpreted together with the realized false-discovery proportion.

Copy number is a gene-level property of A375 and is constant across the plasmid and terminal libraries, so it cannot be identified as a sample-level coefficient. We instead applied MAGeCK 0.5.9.5's official post-fit piecewise correction to both methods' gene effects using the A375 profile from DepMap Public 19Q3 [35, 36]. This changes effect rankings but does not refit either count likelihood.

Supplementary Table S3: Sanson A375 endpoint sensitivity analysis. Recall and gold-set FDP use nominal FDR 0.05 and a negative effect. The last column is Spearman correlation between effect and copy number among reference nonessential genes. No maximum is highlighted because the ranking confidence intervals overlap and the gold-standard set is enriched for positives.

Method	AUROC	Avg. precision	Recall	Gold-set FDP	CNV corr.
■ BARCS	0.9598	0.9786	0.7231	0.0118	−0.185
■ BARCS + CNV	0.9533	0.9762	0.7231	0.0118	−0.042
■ Official MAGeCK-MLE	0.9598	0.9795	0.7098	0.0074	−0.207
■ Official MAGeCK-MLE + CNV	0.9501	0.9762	0.7098	0.0074	−0.006

CNV correction reduces the intended nuisance association, from −0.185 to −0.042 for BARCS and from −0.207 to −0.006 for MAGeCK-MLE, but decreases AUROC for both methods. After correction, the BARCS–MAGeCK AUROC difference is 0.00316 (paired stratified-bootstrap 95% CI −0.00265 to 0.00924), and the average-precision difference is effectively zero (95% CI −0.00349 to 0.00295). At nominal FDR 0.05, BARCS has 1.33 percentage points more recall but also a larger realized FDP within the gold-standard set (0.0118 versus 0.0074). These data do not establish a ranking or calibration advantage for either count model.